

THERMAL ENGINEERING – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.

Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.

Answer five questions from Group-A & B, as directed.

1. A. Answer the following questions (any ten): 10x1
- i) Which of the following is a non-renewable resource? – (a) Coal (b) Forests (c) Water (d) Wildlife.
 - ii) Which of the following is a disadvantage of most of the renewable energy sources? – (a) Highly polluting (b) High waste disposal cost (c) Unreliable supply (d) High running cost.
 - iii) Both power and manure is provided by – (a) Nuclear plants (b) Thermal plants (c) Biogas plants (d) Hydroelectric plant.
 - iv) The amount of heat required to raise the temperature of the unit mass of gas through one degree at constant volume, is called – (a) specific heat at constant volume (b) specific heat at constant pressure (c) kilo Joule (d) none of these.
 - v) An adiabatic process is one in which – (a) no heat enters or leaves the gas (b) the temperature of the gas changes (c) the change in internal energy is equal to the mechanical work done (d) all of the above.
 - vi) Work done in a free expansion process is – (a) zero (b) minimum (c) maximum (d) positive.
 - vii) The ratio of specific heat at constant pressure (C_p) and specific heat at constant volume (C_v) is – (a) equal to one (b) less than one (c) greater than one (d) none of these.
 - viii) The value of gas constant (R) in S.I. units is – (a) 0.287 J/kgK (b) 2.87 J/kgK (c) 28.7 J/kgK (d) 287 J/kg.
 - ix) With usual notation $\Delta x/kA$ is called – (a) Thermal resistance (b) Thermal conductance (c) Heat flux density (d) Thermal loading.
 - x) Pressure for Water at critical point – (a) 221.2 bar (b) 221.2 N/sq.m (c) 225.65 kgf/cm² (d) None of the above.
 - xi) The sum of internal energy (U) and the product of pressure and volume ($p.v$) is known as – (a) workdone (b) entropy (c) enthalpy (d) none of these.
 - xii) The boiling point of steam depends on – (a) Temperature (b) pressure (c) Enthalpy (d) all of the above.

B. State whether the following statements are True or False (any ten):

10x1

- i) In the first law of thermodynamics, the total energy of the system remains constant.
- ii) The kinetic energy of molecules of a gas becomes zero at absolute zero temperature.
- iii) Entropy is a point function.
- iv) The solid fuels have higher calorific value than liquid fuels.
- v) At the time of condensation heat released by steam is sensible heat.
- vi) Steam is a non homogeneous compound.
- vii) 2nd law of thermodynamics indicates the direction of heat transfer.
- viii) A black body reflects all the radiation energy.
- ix) PMM of the second kind is always possible to construct.
- x) Work done is a path function.
- xi) At triple point only steam exists.

Group-A

Answer any three questions.

- 2. Classify energy sources. Discuss disadvantages of Renewable energy. Write short notes on Geothermal Energy, Fossil Fuel, Green Energy. 2+2+6
- 3. Discuss Zeroth law of Thermodynamics. Derive Steady flow energy equation. Write down short notes on Intensive and Extensive properties, quasi static process. 2+4+4
- 4. A non flow Quasi static process (Reversible) occurs for which $p = (-3V+16)$ bar, where V is volume in m^3 . What is the work done if V changes from 2 to 6 m^3 . Derive $PV^n = \text{Const.}$ (with usual notation) 4+6
- 5. Define heat engine, Refrigerator, Heat pump. Prove Reversible engines working between two fixed temperature always be more efficient than any other engine working between same two fixed temperatures. 6+4
- 6. 0.45 kg of Air at 179°C expands adiabatically and reversibly to three times its original volume and during the process there is a fall in temperature to 15.5°C. The work done during the process is 53 kJ. Calculate C_p and C_v . Derive the expression of Enthalpy for constant pressure non flow process. 7+3

Group-B

Answer any two questions.

- 7. Define Dryness fraction, Triple point, Critical point, Super heated steam. Discuss phase transformation with diagram. (4x2)+2
- 8. Find the enthalpy, Internal Energy, and Entropy of 1 kg of steam at a pressure of 10 bar when steam is (a) Dry saturated, (b) When steam is .75 Dry, (c) When steam is at 250°C. Use steam table. 10
- 9. Discuss Fourier's Law of Conduction. Derive an expression of conduction through a flat composite wall of several homogeneous layers. 3+7
- 10. Discuss cross flow heat exchanger. Define Grey body, Black body, Absorptivity, Stefan Boltzman law. 2+(4x2)